

Madhavan Balaji

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PROFESSIONAL SUMMARY

AI/ML Engineer and MS CS Graduate from Indiana University building production **RAG pipelines**, **multi-agent AI systems** (LangGraph, CrewAI), and full-stack NLP applications. Skilled at deploying scalable ML services via **FastAPI**, **Docker**, and **CI/CD** and wiring LLMs (Anthropic, HuggingFace, Ollama) into reliable, observable production systems. Passionate about turning unstructured data into actionable intelligence that users can trust.

TECHNICAL SKILLS

NLP & LLMs: LangChain, LangGraph, CrewAI, HuggingFace Transformers, OpenAI APIs, RAG Pipelines, LlamaIndex, ChromaDB, Text Classification, Entity Extraction, Document Understanding, PEFT/LoRA, Semantic Search

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Vision Transformers (ViT), Stable Diffusion, FAISS, Qdrant, Sentence Transformers

Backend & Infra: Python, FastAPI, RESTful APIs, Docker, Kubernetes, CI/CD (GitHub Actions), Git, Slurm, React

Cloud & Data: GCP, AWS, Render Cloud; Pandas, NumPy, SQL; Model Evaluation, Inference Optimization, Observability

EXPERIENCE

Technical Instructor & Curriculum Lead

Aug 2025 – Present

Indiana University Indianapolis

Indianapolis, IN

- Redesigned Algorithm workshops into visual-first, modular sessions for 30+ students, cutting clarification queries by **40%** and lifting assessment outcomes.
- Ran 1:1 debugging and code-refactoring sessions modeled on DevRel feedback loops, steering students toward production-quality design patterns.
- Launched a competitive coding leaderboard and peer-review program, growing consistent participation across a 30+ member technical community.

Data Analyst Intern

Sep 2022 – Nov 2022

Coincent

Bangalore, India

- Engineered a **CNN-based fashion retrieval system** hitting **90% classification accuracy**, converting unstructured image data into a fast, queryable product search engine.
- Profiled and optimized the feature extraction pipeline with image processing techniques, cutting inference latency by **30%** and unlocking real-time throughput.
- Trained a multi-class classifier across **1,000+ clothing categories**, powering personalized recommendations that drove a **40% lift in user engagement**.

KEY AI PROJECTS

Sovereign SRE – Autonomous Self-Healing Infrastructure | [GitHub](#) | [Live App](#)

May 2026

- Architected an autonomous SRE system wiring a **LangGraph** state machine to a **CrewAI** multi-agent pipeline (Senior SRE + System Researcher) backed by **LlamaIndex** + **ChromaDB** RAG, enabling end-to-end incident detection, root cause analysis, and AI-generated patch creation without human toil.
- Integrated **Ollama (Llama 3.3)**, **Anthropic**, and **Gemini** APIs with Human-in-the-Loop approval gates and automated **GitHub** PR creation, eliminating manual incident response while keeping humans in control of critical fixes.
- Orchestrated all services via **Docker Compose** with real-time **WebSocket** streaming to a **Next.js** dashboard deployed on **AWS Amplify**, delivering live observability across the full autonomous healing pipeline.

DocuMind – Multi-Agent Research Intelligence Platform | [GitHub](#) | [Live Demo](#)

Jan 2026

- Built a stateful **LangGraph** workflow orchestrating **CrewAI** agents (Researcher, Analyst, Writer, Critic) with Hybrid RAG combining dense **Qdrant** search, sparse **BM25**, RRF fusion, and **CrossEncoder** reranking, achieving **167 texts/sec** throughput and **sub-15ms** retrieval at p50.
- Designed an **asyncpg/PostgreSQL** persistence layer with a **FastAPI** REST + **WebSocket** API and **Streamlit** dashboard, enabling full session audit trails with real-time agent reasoning visibility.
- Shipped a custom **MCP server** with 6 registered tools deployed on **AWS**, enabling direct research queries from Claude Desktop and eliminating context-switching between AI and knowledge workflows.

DentaVision – Vision Transformer SaaS | [GitHub](#) | [Live App](#)

May 2025

- Fine-tuned a pretrained **ViT** (timm) for multi-class dental classification served via **FastAPI** with GPU/CPU auto-detection, achieving **92.4% top-1 accuracy** (+8.1 pts over CNN baseline) at **P95 latency of 310ms**.
- Built a **React 18** + **TypeScript** + **Tailwind CSS** frontend that converts model logits into structured clinical reports, reducing time-to-first-prediction to **1.2s** – a **41%** improvement over the prior offline workflow.
- Containerized backend and frontend separately in **Docker** with stateless inference and **Render**-ready configs, achieving zero-persistence production deployment across heterogeneous compute environments.

EDUCATION

Indiana University Indianapolis

Exp. May 2026

Master of Science in Computer Science

Indianapolis, IN

- Coursework:** Deep Learning, Machine Learning, Computer Vision, Algorithms, Data Mining, Database Design

Sri Venkateswara College of Engineering

2024

Bachelor of Information Technology

Chennai, India

PATENT

Published: Caremate – An Integrated Elderly Support System | [View Document](#)